

Dr. Samik Mitra

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Research Summary

Relativistic astrophysicist whose research connects the physics of black hole accretion environments to the broader observable signatures they produce – from magnetized plasma dynamics to gravitational-wave and particle-physics fingerprints. Combines semi-analytical (steady-state, post-Newtonian) modeling with GRMHD simulation and general relativistic ray-tracing to link accretion theory with observed polarization, QPO, and radio/X-ray signatures, and extends this framework to gravitational-wave damping in turbulent disks and indirect dark-matter constraints from galaxy-cluster environments. Author of 8 refereed papers (4 as first author) with 5 further papers under review; invited speaker at Harvard CfA, Fudan, SHAO, and IISc. Currently a postdoctoral fellow at ICTS–TIFR, building an independent research program that treats accretion physics, gravitational-wave phenomenology, and dark-matter searches as different observational windows onto the same compact-object environments.

Research Experience / Collaborative Visits

Postdoctoral Fellow, ICTS–TIFR, India <i>Mentor: Dr. Pallavi Bhat</i>	April 2025 – Present
Visitor, Niels Bohr Institute, University of Copenhagen <i>Host: Prof. Martin E. Pessah</i>	June 2026
Visitor, Nicolaus Copernicus Astronomical Centre, Warsaw <i>Host: Prof. Włodek Kluzniak</i>	May – June 2026
Visitor, Czech Academy of Sciences, Prague <i>Host: Dr. Jiri Horak</i>	May 2026
Visitor, Masaryk University, Brno <i>Host: Dr. Michal Zajaček</i>	May 2026
Visiting Research Scholar, Shanghai Astronomical Observatory (SHAO), China <i>Host: Prof. Fu-Guo Xie</i>	Nov 2024 – Jan 2025
Visiting Research Scholar, IIT Indore, India <i>Hosts: Prof. Bhargav Vaidya & Dr. Deepika Bollimpalli</i>	October 2024

Ongoing Projects

- Quasi-normal mode excitations originating from turbulent accretion disks.
- Effect of turbulent magnetic fields on horizon-scale images.
- Theoretical model development of the ‘fundamental plane’ to explain X-ray to radio observations.
- Damping of gravitational waves while passing through a viscous medium.

Education

Ph.D. in Astrophysics, IIT Guwahati, India • Thesis: <i>Magnetohydrodynamic and Hydrodynamic Studies of Relativistic Accretion Flows around Black Holes</i> • Thesis Advisor: Prof. Santabrata Das • Ph.D. viva-voce: February 6, 2025 Thesis submitted: October 1, 2024 • Prime Minister’s Research Fellow (PMRF), August 2020 – July 2024 • Topped Ph.D. coursework with 9.69/10 (July 2020) • Ph.D. mini-project: <i>Dissipative accretion flows around black holes</i> (June 2020)	July 2019 – Feb 2025
M.Sc. in Astrophysics, University of Delhi, India • Department of Physics & Astrophysics; aggregate 64.40% (First Class)	July 2016 – June 2018

- Specialization in Theoretical Physics (Astrophysics, General Relativity, Cosmology, Plasma Physics, Quantum Field Theory); ranked among top 18 of 294 students
- M.Sc. project: *Astrophotography of a Cluster (Orion Field) using a CCD Camera*, under Prof. H. P. Singh

B.Sc. (Honours) in Physics, University of Calcutta, India

July 2013 – June 2016

- Raja Peary Mohan College; aggregate 66.25% (First Class); ancillary subjects: Mathematics, Chemistry

Secondary & Higher Secondary, Kanailal Vidyamandir, West Bengal

July 2005 – May 2013

- Class XII (W.B.C.H.S.E.), Science stream (Physics-Chemistry-Mathematics-Biology): 81%; top 1% in West Bengal board (2013)
- Class X (W.B.B.S.E.): 87.5% (2011)

Research Interests

Primary: Relativistic hydrodynamics and magnetohydrodynamics, accretion flows, relativistic jets, shock waves, large-scale dynamos and reconnection, quasi-periodic oscillations (QPOs), radiative processes, state transitions in black hole X-ray binaries (BH-XRBs), polarized emission.

Additional: Numerical relativity, post-Newtonian theory.

Publications

As of 5 July 2026: **8** refereed papers (**4** first-author, **4** co-authored) and **1** published conference proceeding. **5** additional papers under review (**2** first-author, **3** co-authored), plus **2** conference proceedings submitted. Full list via [ADS](#) (ORCID: 0000-0002-5730-0376).

First-Author Publications

1. **Mitra S.**, Vrba J., Rayimbaev J., Stuchlík Z., Ahmedov B., “Charged particles and quasi-periodic oscillations in Black-bounce-Reissner-Nordström geometry in braneworlds,” *Phys. Dark Universe* **46** (2024).
2. **Mitra S.**, Das S., “Low angular momentum general relativistic magnetohydrodynamic accretion flow around rotating black holes with shocks,” *ApJ* **971**, 1 (2024).
3. **Mitra S.**, Ghoreyshi S. M., Mosallanezhad A., Abbassi S., Das S., “Global transonic hot accretion flow with thermal conduction,” *MNRAS* **523**, 4431–4440 (2023).
4. **Mitra S.**, Maity D., Dihingia I. K., Das S., “Study of general relativistic magnetohydrodynamics accretion flows around black holes,” *MNRAS* **516**, 5091–5109 (2022).

Co-Authored Publications

1. Turimov B., Karshiboev K., Abdujabbarov A., **Mitra S.**, Karshiboev S., “Orbital precession in Janis–Newman–Winicour spacetime,” *Galaxies* **12** (2024).
2. Ghorani E., **Mitra S.**, Rayimbaev J., et al., “Constraints on Einstein–Geometric Proca–AdS black hole from QPO data,” *EPJC* **84** (2024).
3. Nazari E., **Mitra S.**, Abbassi S., Das S., “Accretion flows around spinning compact objects in the post-Newtonian gravity,” *JCAP* **05**, 117 (2024).
4. Das S., Nandi A., Stalin C. S., Rakshit S., Dihingia I. K., Singh S., Aktar R., **Mitra S.**, “On the origin of core radio emissions from the black hole sources in the realm of relativistic shocked accretion flow,” *MNRAS* **514**, 1940–1951 (2022).

Submitted

1. **Mitra S.**, Czerny B., Zajaček M., et al., “Low-angular-momentum accretion shocks as a sufficient energy reservoir for Sgr A* X-ray flares,” submitted to *ApJL* (2026).
2. **Mitra S.**, Santra R., “Connecting BCG X-ray Cavities to Finite-angular-momentum GRMHD Inflows – I,” submitted to *ApJ* (2026).
3. Zajaček M., Czerny B., Mondek M., **Mitra S.**, et al., “The Galactic Center as a connected ecosystem across spatial and temporal scales,” submitted to *Nature Astronomy* (2026).
4. Santra R., **Mitra S.**, “Radio synchrotron limits on WIMP annihilation in the Frontier Fields cluster Abell 2744,” submitted to *JHEAP* (2026).
5. Santra R., **Mitra S.**, “Broadband constraints on axion-like particle dark matter from the Coma cluster radio halo and prospects for the SKA,” submitted to *PRD* (2026).

Conference Proceedings: Published

1. **Mitra S.**, Dihingia I. K., Das S., Maity D., “Properties of general relativistic magneto-hydrodynamics (GRMHD) accretion flows around black holes,” 44th COSPAR Assembly (2022).
2. **Mitra S.**, “Properties of low angular momentum general relativistic MHD flows around black holes,” 17th Marcel Grossmann Meeting (2024).

Conference Proceedings: Submitted

1. **Mitra S.**, Chen Y., Bhat P., Cardoso V., “Gravitational wave emission from turbulent accretion disks around black holes,” IAU Symposium 405 (2026).
2. **Mitra S.**, Czerny B., Zajaček M., “Nature of NIR and MIR flares in Sgr A*,” IAU Symposium 405 (2026).

Invited Talks & Seminars

- “Gravitational wave emission from turbulent accretion disks around black holes,” CAMK, Warsaw (June 2026).
- “Importance of low angular momentum magnetized accretion flows around Sgr A*,” Czech Academy of Sciences, Prague (May 2026).
- “Magnetized accretion flows around rotating black holes,” Silesian University, Opava (May 2026).
- “A hybrid framework for magnetohydrodynamic accretion disks: Analytical and simulation perspectives,” IUCAA, India (Jan 2026).
- “Overview of magnetized accretion flows around black holes,” IISc Bangalore, India (Aug 2025), invited by Dr. Prantika Bhowmick.
- “Impact of non-zero net vertical magnetic field on thick accretion torus around black holes,” 5th China–India Workshop on High Energy Astrophysics (Dec 2024), invited by Prof. Cosimo Bambi.
- “Standing GRMHD shocks in accretion flows around rotating black holes,” TDLI, Shanghai (Dec 2024).
- “An overview of the magnetized advective flows around black holes,” SWIFAR, Yunnan University, China (Dec 2024), invited by Prof. Chandra Bahadur Singh.
- “Study of magnetized accretion flows around black holes: A general relativistic approach,” CfAA, Fudan University, China (Dec 2024), invited by Prof. Cosimo Bambi.
- “Properties of low angular momentum GRMHD accretion flows around rotating black holes,” Center for Astrophysics, Harvard University, USA (Nov 2024), invited by Dr. D. C. Palumbo.
- “Study of magnetized accretion flows around black holes: A general relativistic approach,” SHAO, Chinese Academy of Sciences (Nov 2024), invited by Prof. Fu-Guo Xie.
- “Properties of low angular momentum GRMHD accretion flows around black holes,” DAASE, IIT Indore (Oct 2024), invited by Dr. Bhargav Vaidya.
- “Properties of general relativistic magnetohydrodynamics accretion flows around black holes in steady-state limit,” Research Center for Astronomy and Applied Mathematics, Academy of Athens, Greece (Jul 2022), invited by Prof. I. Contopoulos.

Conference Talks & Posters Presented

- “Impact of non-zero net vertical magnetic field on thick accretion torus around black holes,” RAGtime 26 (Dec 2024), invited by Dr. Jiri Horak [Online].
- “Properties of low angular momentum GRMHD accretion flows around black holes” (oral) and “Accretion flows around spinning compact objects in the post-Newtonian regime” (poster), Galactic and Extragalactic X-ray Transients, Warsaw, Poland (Sep 2024).
- “Properties of low angular momentum general relativistic MHD flows around black holes” and “Accretion flows around spinning compact objects in the post-Newtonian regime,” 17th Marcel Grossmann Meeting, Pescara, Italy (Jul 2024) [Online].
- “Impact of net vertical magnetic field on thick accretion torus around black holes,” 10th International Conference on Gravitation and Cosmology (ICGC 2023), IIT Guwahati, India (Dec 2023) – also hosted.
- “Properties of low angular momentum trans-magneto-sonic accretion flows around black holes,” RAGtime 25 (Nov 2023), invited by Prof. Martin Kološ [Online].
- “Shock induced GRMHD accretion flows around black holes,” Challenges in Computational Astrophysics (ChalCA V) (Oct 2023) [Online].
- “Global structure of shock-induced general relativistic magnetohydrodynamic (GRMHD) accretion flows around black holes,” RETCO-V, Kodaikanal Solar Observatory, IIA, India (Apr 2023).
- “Global structure of general relativistic magneto-hydrodynamics accretion flows around black holes,” Young Astronomer’s Meet 2022, ARIES, Nainital, India (Nov 2022).

- “Properties of General Relativistic Magnetohydrodynamic (GRMHD) accretion flows around black holes,” 44th COSPAR Assembly, Athens, Greece (Jul 2022).
- “Steady state general relativistic magnetohydrodynamic accretion flows around black holes,” ASI 2022 (poster) (Mar 2022) [Online].
- “Steady state general relativistic magnetohydrodynamic accretion flows around black holes” (oral) and 3-Minute Thesis (3MT), Research and Industrial Conclave 2022, IIT Guwahati (Jan 2022) – *2nd Prize, 3MT* [Online].
- “Steady-state analysis of General Relativistic Magnetohydrodynamics (GRMHD) accretion flows around black hole space-time,” XII Biennial National Conference of PANE (Dec 2021) – *Best Oral Presentation* [Online].

Conferences, Schools & Workshops Attended

- International School, ‘The Future of Gravitational-Wave Astronomy,’ ICTS, Bengaluru (Oct 2025) – also served as observer.
- International School, ‘Summer School on Gravitational Wave Astronomy,’ ICTS, Bengaluru (Jul 2025).
- International School, ‘Beyond the Horizon: Testing the Black Hole Paradigm,’ ICTS, Bengaluru (Mar–Apr 2025).
- International Conference on Black Holes Inside and Out (BHIO 2024), Niels Bohr Institute, Copenhagen (Aug 2024) [Online].
- ‘Regular black holes in quantum gravity and beyond: from theory to shadow observations’ (Oct 2022) [Online].
- School on Black Holes and Gravitational Waves, IIT Madras (Sep 2022) [Online].
- International School for Space Simulations (Sep 2022) [Online].
- IAGR School, IIT Gandhinagar (May 2022) [Online].
- Testing Aspects of General Relativity, IIT Gandhinagar (Mar 2022) [Online].
- Black Hole Astrophysics with VLBI, Institute of Cosmic Ray Research, University of Tokyo (Feb 2022) [Online].
- International Conference on Black Holes Inside and Out (BHIO 2021) (Sep–Oct 2021) [Online].
- Workshop on Chandra’s Contribution in Plasma Astrophysics, School of Plasma Physics, JNU (Oct 2021) [Online].
- Online Workshop on HPC in Astrophysics and Astronomy, NCRA-TIFR & IIT Kharagpur (Sep 2021) [Online].
- Prof. S.M. Chitre Memorial Symposium on Frontiers of Astrophysics and Fluid Dynamics, UM-DAE CEBS, University of Mumbai (Jul 2021) [Online].
- International Workshop on Numerical Astrophysics, Ferdowsi University of Mashhad, Iran (Jul 2020) [Online].

Awards & Honors

- Qualified Graduate Aptitude Test in Engineering (GATE), MHRD, Govt. of India (2019).
- Prime Minister’s Research Fellowship (PMRF), MHRD, Govt. of India (Sept 2020 – Jul 2024); fellowship: EUR 40,312; contingency grant: EUR 8,810 over 4 years.
- PANE Young Researcher Award, Astrophysics & Cosmology category (Dec 2021).
- 2nd Place, Three Minute Thesis (3MT) Presentation, IIT Guwahati RIC (2022).
- Featured as Commendable Research by PMRFs, Ministry of Education, PMRF, Govt. of India.
- Received postdoctoral offers from Tartu Observatory, Zhejiang University, and Ningbo University (2024 cycle).

Research Techniques & Software

RMHD/GRMHD Codes: Athena++ (with Dr. Bhupendra Mishra); BHAC (with Dr. Indu K. Dihingia & Dr. Pallavi Bhat); PLUTO (with Prof. Bhargav Vaidya).

Ray-Tracing: RAPTOR (advanced), Kgeo (implemented analytical model with Dr. Andrew Chael), ipole, blacklight, Gyoto (beginner).

High-Performance Computing: PARAM-ISHAN (IIT Guwahati); CONTRA, SONIC, MARIO (ICTS).

Languages & Tools: Fortran, C, Python, Julia, Mathematica, VisIt, ParaView.

Teaching & Mentoring

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- Teaching Assistant, IIT Guwahati – multiple courses across 8 semesters.
 - Teaching Assistant, Department of Physics, North Gauhati College (Gauhati University) – 4 years.
 - Teaching Assistant, ICTS Ph.D. course *Advanced Classical Electrodynamics* (with Pallavi Bhat, Maneesha P., R. Loganayagam).
 - Mentored B.Tech./M.Sc. thesis students: Ashish Menon (now TU Eindhoven), Tanya Dhama (NCBS–ICTS), Vikhyat (IISER Pune), Ayan Chakrabarty (IIT Guwahati), Subhajit Das (IIT Guwahati), Alay Pal, Chinmay.
 - Supervising at ICTS: Aniket Nath (NISER) and Anurag Reddy (IISER Pune; incoming Ph.D. at University of Jena) in GRMHD, GRRT, and Numerical Relativity.

Conference Organization

- Student LOC, North-East Meet of Astronomers (NEMA-VI), IIT Guwahati (Nov 2020, online).
- Student LOC, 10th International Conference on Gravitation and Cosmology (ICGC 2023), IIT Guwahati (Dec 2023).
- SOC, Young Astronomer’s Meet (YAM 2024 & YAM 2025), Christ University & TIFR, India.

Science Outreach

- Founder of a newsletter and page curating and summarizing papers from astro-ph (*Samik’s Newsletter*).
- First science-pub talk ‘Axis-Mundi’ on “What do cosmic beasts look like through computers,” Brno, Czechia.
- Reviewer for *The Astrophysical Journal* and *Physics Letters B*.
- “How a Black Hole Lights Up its Surroundings,” National Science Day, IIT Guwahati (2023).
- “Magnetohydrodynamic flows around black holes in strong gravity regime: an analytical and numerical study,” National Science Day, IIT Guwahati (2022).

References

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